2023 SSMO Accuracy Round Solutions

SMO Team

AR 1: Mr. Sammy proposes a Hamburger Proclamation, which has 500 lines, divided into paragraphs of 5 lines each. It takes him 23 seconds to read each line. Additionally, he adds a 0.5 second pause between two lines in a paragraph, and a 2 second pause between paragraphs. If it takes him S minutes to read the whole Hamburger Proclamation, find $\lfloor 10S \rfloor$.

Answer: $\boxed{1983}$

Solution: There are 100 paragraphs, so there are 499 pauses between all lines and 99 pauses between paragraphs. Then, it takes him

$$23 \cdot 500 + (499 - 99) \cdot 0.5 + 99 \cdot 2 = 11898$$

seconds to read the entire passage. Then, $10S$ is $\frac{10}{60} = \frac{1}{6}$ of this value, so $\lfloor 10S \rfloor = \boxed{1983}$.

AR 2: Suppose that the average of all n-digit palindromes is denoted by P_n and the average of all n-digit numbers is denoted by N_n. Find the value of $\left\lfloor \sum_{n=1}^{100} (P_n - N_n) \right\rfloor$.

Answer: $\boxed{49}$

Solution: We claim that $P_n - N_n = \frac{1}{2}$ for all $n \neq 1$. From linearity of expectation, P_n and N_n are each the sum of the average values of their digits, weighted by place value. The first $n-1$ digits contribute equally to both, while the units digit of P_n equals its leading digit and ranges from 1 to 9, and the units digit of N_n ranges from 0 to 9, giving

$$\frac{1+9}{2} - \frac{0+9}{2} = \frac{1}{2}.$$

For $n = 1$, we have $P_1 = N_1 = 5$, so $P_1 - N_1 = 0$. Thus,

$$\left\lfloor \sum_{n=1}^{100} (P_n - N_n) \right\rfloor = \left\lfloor 99 \cdot \frac{1}{2} \right\rfloor = \boxed{49}.$$

AR 3: Suppose that a, b, c are real numbers such that

$$\begin{aligned} a + b - c &= 4, \\ a^2 + b^2 + c^2 &= 14, \text{ and} \\ a^3 + b^3 - c^3 &= 34. \end{aligned}$$

Find the sum of all possible values of $a + b + c$.

Answer: $\boxed{4}$

Solution: Let $x = -c$. Then

$$\begin{aligned} a + b + x &= 4, \\ a^2 + b^2 + x^2 &= 14, \\ a^3 + b^3 + x^3 &= 34. \end{aligned}$$

Squaring the first equation and subtracting the second gives

$$2(ab + ax + bx) = 2 \implies ab + ax + bx = 1.$$

Multiplying by $a + b + x = 4$, we obtain

$$(a^2b + ab^2 + ax^2 + a^2x + b^2x + bx^2) + 3abx = 4.$$

Cubing $a + b + x = 4$ yields

$$\begin{aligned} (a^3 + b^3 + x^3) + 3(a^2b + ab^2 + ax^2 + a^2x + b^2x + bx^2) + 6abx &= (a + b + x)^3 \\ 34 + 3(4 - 3abx) + 6abx &= 64 \implies \\ 46 - 3abx &= 64 \implies abx = -6. \end{aligned}$$

Thus, a, b, and x are the roots of the polynomial

$$y^3 - 4y^2 + y + 6 = (y - 3)(y - 2)(y + 1).$$

Hence, the possible values of $a + b + c = a + b - x$ are

$$3 + 2 - (-1) = 6, \quad 3 + (-1) - 2 = 0, \quad 2 + (-1) - 3 = -2,$$

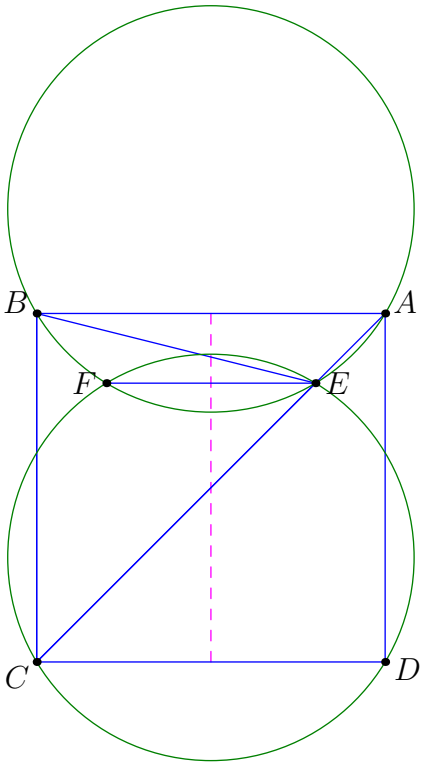
and their sum is

$$\boxed{4}.$$

AR 4: In square $ABCD$, point E is selected on diagonal AC. Let F be the second intersection of the circumcircles of triangles ABE and CDE. Given that $AB = 10$ and $EF = 6$, find the maximum possible area of triangle BEC.

Answer: $\boxed{40}$

Solution:



Since the centers of both circumcircles lie on the perpendicular bisector of AB and CD, their common chord EF is perpendicular to it and bisected by it. So, F is the reflection of E across the perpendicular bisector of CD. Define coordinates $A = (10, 10)$, $B = (0, 10)$, $C = (0, 0)$, and $D = (10, 0)$. Since $EF = 6$, it follows that E is either $(8, 8)$ or $(2, 2)$, so F is either $(2, 8)$ or $(8, 2)$. The first case produces the larger area

$$\frac{1}{2} \cdot 10 \cdot 8 = \boxed{40}.$$

AR 5: Define the "relationship" between two positive integers a and b to be $\frac{\tau(ab)}{\tau(a)\tau(b)}$ where $\tau(x)$ is the number of divisors of x. Find the sum of all integers $1 \leq n \leq 100$ which have a relationship of $\frac{3}{4}$ with 360.

Answer: $\boxed{340}$

Solution: Since $360 = 2^3 \cdot 3^2 \cdot 5$, we have $\tau(360) = 4 \cdot 3 \cdot 2 = 24$. Thus, for any n satisfying the given $\frac{3}{4}$ relationship, we have

$$\frac{\tau(360n)}{24\tau(n)} = \frac{3}{4} \implies \frac{\tau(360n)}{\tau(n)} = 18.$$

Let $n = 2^a \cdot 3^b \cdot 5^c \cdot p$ with $\gcd(p, 360) = 1$. Then,

$$\frac{(a+4)(b+3)(c+2)\tau(p)}{(a+1)(b+1)(c+1)\tau(p)} = 18.$$

Substituting $d = a + 1$, $e = b + 1$, and $f = c + 1$ gives

$$\frac{(d+3)(e+2)(f+1)}{def} = 18.$$

Since $\frac{d+3}{d}$, $\frac{e+2}{e}$, and $\frac{f+1}{f}$ are decreasing, we must have $d = 1$, as otherwise the left hand side is at most $\frac{(2+3)(1+2)(1+1)}{2\cdot1\cdot1} = 15 < 18$. Then,

$$\frac{(e+2)(f+1)}{ef} = \frac{9}{2} \implies 7ef - 2e - 4f - 4 = 0,$$

which factors as $(7e-4)(7f-2) = 36$, yielding $e = 1$ and $f = 2$. Thus $a = 0$, $b = 0$, and $c = 1$, so $n = 5p$ with $\gcd(p, 360) = 1$. Since $n \leq 100$, we have $p \leq 20$, giving $p = 1, 7, 11, 13, 17, 19$. Therefore, the sum of all such n is

$$5(1 + 7 + 11 + 13 + 17 + 19) = \boxed{340}.$$

AR 6: Let the roots of $P(x) = x^3 - 2023x^2 + 2023^{2023}$ be α, β, γ. Find

$$\frac{\alpha^2+\beta^2}{\alpha+\beta} + \frac{\beta^2+\gamma^2}{\beta+\gamma} + \frac{\gamma^2+\alpha^2}{\gamma+\alpha}.$$

Answer: $\boxed{8092}$

Solution: From Vieta's formulas, $\alpha + \beta + \gamma = 2023$ and $\alpha\beta + \alpha\gamma + \beta\gamma = 0$, so

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\alpha\gamma+\beta\gamma) = 2023^2.$$

Hence,

$$\frac{\alpha^2+\beta^2}{\alpha+\beta} = \frac{(\alpha^2+\beta^2+\gamma^2)-\gamma^2}{(\alpha+\beta+\gamma)-\gamma} = \frac{2023^2-\gamma^2}{2023-\gamma} = 2023+\gamma,$$

and similarly

$$\frac{\alpha^2+\gamma^2}{\alpha+\gamma} = 2023+\beta \quad \text{and} \quad \frac{\beta^2+\gamma^2}{\beta+\gamma} = 2023+\alpha.$$

Therefore,

$$\begin{aligned}\frac{\alpha^2+\beta^2}{\alpha+\beta} + \frac{\beta^2+\gamma^2}{\beta+\gamma} + \frac{\gamma^2+\alpha^2}{\gamma+\alpha} &= (2023+\alpha)+(2023+\beta)+(2023+\gamma) \\ &= 6069 + (\alpha+\beta+\gamma) = 6069 + 2023 = \boxed{8092}.\end{aligned}$$

AR 7: Concentric circles ω and ω_1 are drawn, with radii 3 and 5, respectively. Chords AB and CD of ω_1 are both tangent to ω, and lines AB and CD intersect at P. If $PA = PC = 3$, then the sum of all possible distinct values of $[PAD]$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

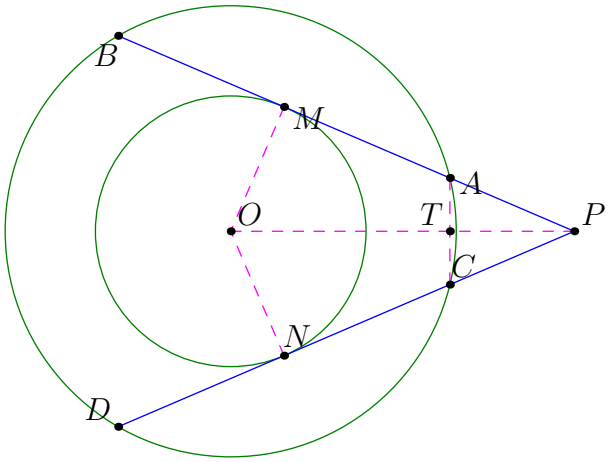
Answer: $\boxed{506}$

Solution: Let O be the common center, and let M and N be the points where AB and CD are tangent to ω, respectively. There are two cases. First assume that P lies outside ω_1. By the Pythagorean Theorem, we have $AB = CD = 2\sqrt{5^2-3^2} = 8$, so $PN = PC + CN = 7$ and $OP = \sqrt{3^2+7^2} = \sqrt{58}$. Let T be the midpoint of CA, which lies on OP by symmetry. Then, $\triangle NOP \sim \triangle TCP$, so since $[NOP] = \frac{1}{2}\cdot 3 \cdot 7 = \frac{21}{2}$, it follows that

$$[CTP] = [NOP]\left(\frac{CP}{OP}\right)^2 = \frac{21}{2}\cdot\frac{9}{58} = \frac{189}{116}.$$

Hence $[PAC] = 2[CTP] = \frac{189}{58}$. Noting that $PC = 3$ and $PD = 3 + 8 = 11$, we obtain

$$[PAD] = \frac{PD}{PC}\cdot[PAC] = \frac{11}{3}\cdot\frac{189}{58} = \frac{693}{58}.$$

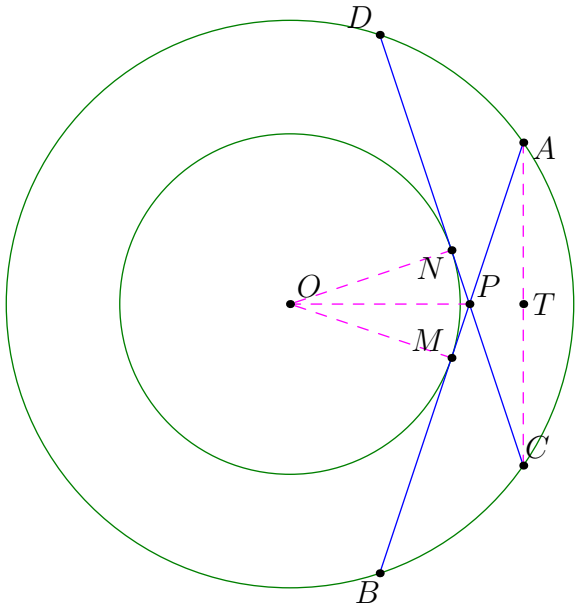


Now consider the case where P lies inside ω_1. We have $PM = PN = 1$ and $OP = \sqrt{10}$, and once again $\triangle NOP \sim \triangle TCP$. Thus,

$$[PAC] = 2[CTP] = \frac{27}{10} \quad \text{and} \quad [PAD] = \frac{9}{2}.$$

The sum of the areas is therefore

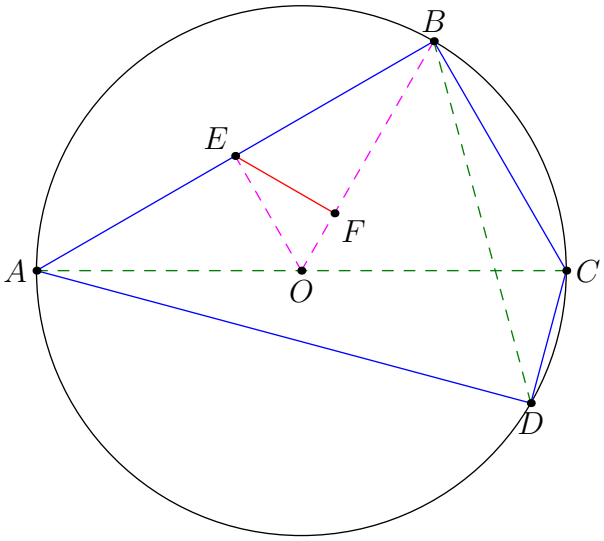
$$\frac{693}{58} + \frac{9}{2} = \frac{477}{29} \implies 477 + 29 = \boxed{506}.$$



AR 8: There is a quadrilateral $ABCD$ inscribed in a circle ω with center O. In quadrilateral $ABCD$, diagonal AC is a diameter of the circle, $\angle BAC = 30°$, and $\angle DAC = 15°$. Let E be the foot of the perpendicular from O to side BA. Let F be the foot of the perpendicular from E to BO. Given that $EF = 3$, the product of the lengths of the diagonals of $ABCD$ can be expressed as $a\sqrt{b}$, where a and b are positive integers and b is not divisible by the square of any prime. Find $a + b$.

Answer: $\boxed{98}$

Solution:



Since $EF = 3$ and $\triangle EOF$ is a 30-60-90 triangle, we have $EO = 2\sqrt{3}$, so $BC = 4\sqrt{3}$ and $AC = 8\sqrt{3}$. Thus the radius of ω is $4\sqrt{3}$, and since $\angle BAD = 30^\circ + 15^\circ = 45^\circ$, the central angle $\angle BOD$ is a right angle, so it follows that $BD = 4\sqrt{6}$. Hence,

$$AC \cdot BD = 8\sqrt{3} \cdot 4\sqrt{6} = 96\sqrt{2},$$

and the answer is $96 + 2 = \boxed{98}$.

AR 9: Consider a 2023×2023 grid called $A = P_{2023}$. We take one of the four smaller 2022×2022 grids located in P_{2023} as P_{2022}. We repeat the process of taking smaller grids until we eventually end at the unit square P_1.

Of the 4^{2022} distinct tuples of shrinking grids $(P_{2023}, P_{2022}, \ldots, P_1)$, let T be the number of these tuples such that their last element is the center square of the original grid A. Find the largest integer a such that $2^a \mid T$.

Answer: $\boxed{16}$

Solution: Let $a_{i,j,k}$ denote the number of tuples with last element at position $0 \le i, j \le k$ when $A = P_{k+1}$, and let $a_{i,j,k} = 0$ if i or j is out of range. We seek $a_{1011,1011,2022}$. We claim that

$$a_{i,j,k} = \binom{k}{i}\binom{k}{j}.$$

From the four possible placements of P_k, we have

$$a_{i,j,k} = a_{i-1,j-1,k-1} + a_{i-1,j,k-1} + a_{i,j-1,k-1} + a_{i,j,k-1},$$

and the closed form satisfies the same recursion since

$$\binom{k}{i}\binom{k}{j} = \left(\binom{k-1}{i-1} + \binom{k-1}{i}\right)\left(\binom{k-1}{j-1} + \binom{k-1}{j}\right).$$

The base cases agree (values vanish out of range and $a_{0,0,0} = 1$), so the formula holds. Thus,

$$a_{1011,1011,2022} = \binom{2022}{1011}^2,$$

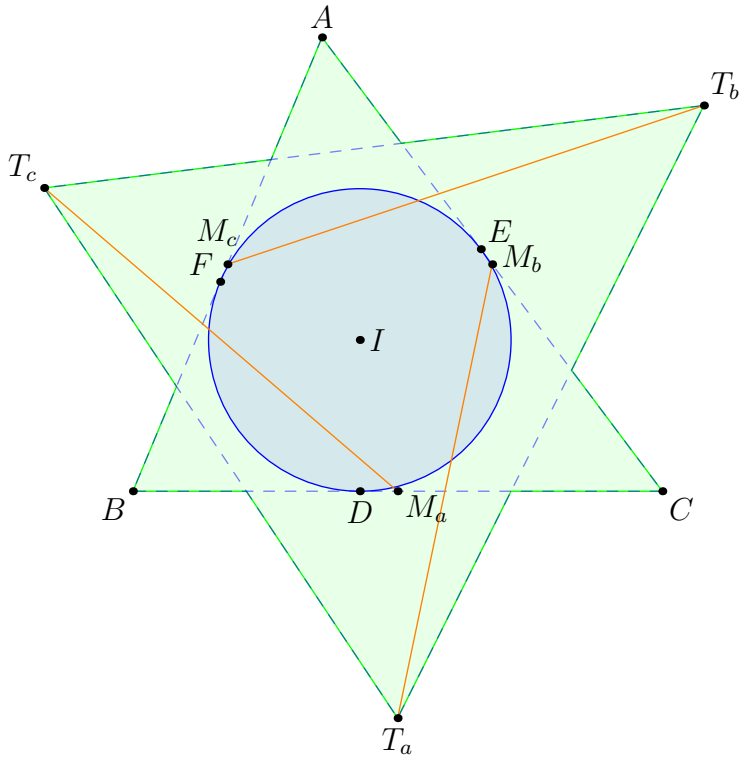
and it suffices to compute $2v_2\left(\binom{2022}{1011}\right)$. From Legendre's formula, $v_2(n!) = n - s_2(n)$, where $s_2(n)$ is the sum of the binary digits of n. Since $1011 = 1111110011_2$ and $s_2(2022) = s_2(1011)$, we have

$$\begin{aligned} 2v_2\left(\binom{2022}{1011}\right) &= 2\big(v_2(2022!) - 2v_2(1011!)\big) \\ &= 2\big(2s_2(1011) - s_2(2022)\big) \\ &= 2s_2(1011) = 2 \cdot 8 = \boxed{16}. \end{aligned}$$

AR 10: Let $\triangle ABC$ be a triangle such that $AB = 13$, $BC = 14$, $CA = 15$. Let the incircle of $\triangle ABC$ touch BC at D, AC at E, and AB at F. Let ℓ_A be the line through the midpoints of AE and AF. Define ℓ_B and ℓ_C similarly. The area of the star created by the union of $\triangle ABC$ and the triangle bounded by ℓ_A, ℓ_B, and ℓ_C can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{2389}$

Solution:



First, note that $AE = AF = 7$, $BD = BF = 6$, $CE = CD = 8$, the inradius is 4, and the area of ABC is 84. Since the midpoints of AE and AF have equal power with respect to the point-circle at A and the incircle, ℓ_A is the radical axis of these two circles, and similarly for ℓ_B and ℓ_C. Let M_a be the midpoint of BC and let T_a be the intersection of ℓ_B and ℓ_C. Then, T_a is the radical center of the point-circles at B and C and the incircle, so T_a lies on the perpendicular bisector of BC and

$$T_aB^2 = T_aC^2 = T_aI^2 - ID^2.$$

Since $M_aD = \frac{BC}{2} - BD = 1$, applying the Pythagorean Theorem to T_aB and T_aI gives

$$M_aT_a^2 + 7^2 = (M_aT_a + 4)^2 + 1^2 - 4^2 \implies M_aT_a = 6.$$

Similarly, with T_b and T_c defined in the same manner,

$$M_bT_b^2 + \left(\frac{15}{2}\right)^2 = (M_bT_b + 4)^2 + \left(\frac{1}{2}\right)^2 - 4^2 \implies M_bT_b = 7,$$

and

$$M_cT_c^2 + \left(\frac{13}{2}\right)^2 = (M_cT_c + 4)^2 + \left(\frac{1}{2}\right)^2 - 4^2 \implies M_cT_c = \frac{21}{4}.$$

Since ℓ_B and ℓ_C meet BC at the midpoints of BD and CD, the cap of the star outside side BC is a triangle with base $\frac{BC}{2}$ and height M_aT_a, and similarly for the other two caps. Therefore, the total area is $[ABC]$ plus the three triangular caps, which is

$$84 + \tfrac{1}{2}\left(6 \cdot 7 + 7 \cdot \tfrac{15}{2} + \tfrac{21}{4} \cdot \tfrac{13}{2}\right) = \frac{2373}{16},$$

and the requested sum is $2373 + 16 = \boxed{2389}$.